CLOSED IMMERSIONS INTO A MCQUILLAN AFFINE FORMAL ALGEBRAIC SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one surjection between limits; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A closed immersion of formal algebraic spaces, and the shape of its base changes to the spectra of quotients. Same chapter, Definition definition-closed-immersion, invoked at proof lines 6149-6151. Makes each base change a closed subscheme of an affine scheme, hence affine, with a surjective ring map.
- The criterion for a McQuillan affine formal algebraic space. Same chapter, Lemma lemma-mcquillan-affine-formal-algebraic-space, invoked at proof lines 6158-6160. Gives that the source is McQuillan once the structure maps of the limit are surjective.
- A representable affine morphism gives a taut ring map. Same chapter, Lemma lemma-representable-affine, invoked at proof lines 6161-6163. Gives tautness of the ring map.

Source reference. Original lemma 0AIC: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2135–2135. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN5: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4239–4239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 03XZ: sources/stacks-chow/tags.txt, source lines 5166–5166. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Definition 0.1. Let S be a scheme. Let $f : Y \to X$ be a morphism of formal algebraic spaces over S. We say f is a *closed immersion* if f is representable by algebraic spaces and a closed immersion in the sense of Bootstrap, Definition 03XZ.

Lemma 0.2. *Let S be a scheme. Let X be a McQuillan affine formal algebraic space over S. Let $f : Y \to X$ be a closed immersion of formal algebraic spaces over S. Then Y is a McQuillan affine formal algebraic space and f corresponds to a continuous homomorphism $A \to B$ of weakly admissible topological S-algebras which is taut, has closed kernel, and has dense image.*

Proof. Write $X = \mathrm{Spf}(A)$ where A is a weakly admissible topological ring. Let I_λ be a fundamental system of weakly admissible ideals of definition in A. Then $Y \times_X \mathrm{Spec}(A/I_\lambda)$ is a closed subscheme of $\mathrm{Spec}(A/I_\lambda)$ and hence affine (Definition 0.1). Say $Y \times_X \mathrm{Spec}(A/I_\lambda) = \mathrm{Spec}(B_\lambda)$. The ring map $A/I_\lambda \to B_\lambda$ is surjective. Hence the projections

$$B = \lim B_\lambda \longrightarrow B_\lambda$$

are surjective as the compositions $A \to B \to B_\lambda$ are surjective. It follows that Y is McQuillan by Lemma 0AIC. The ring map $A \to B$ is taut by Lemma 0AN5. The kernel is closed because B is complete and $A \to B$ is continuous. Finally, as $A \to B_\lambda$ is surjective for all λ we see that the image of A in B is dense. $\square$